

Structured Query Language SQL

Les squelettes du SELECT

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« Squelette » de requêtes

squelette 1 - Projection : SELECT attributs FROM table

squelette 2 - Implication de plusieurs entités : SELECT attributs FROM table1 join table2

squelette 3-1 - Restriction simple : SELECT attributs FROM tables WHERE conditions

squelette 3-2 - Restriction complexe : SELECT attributs FROM tables
WHERE conditions avec sous-requête

squelette 4 : Fonction de groupes ou d'agrégation : SELECT attributs, fonction(attribut)
FROM tables GROUP BY ...

squelette 5-2 : Agrégation conditionnée par une constante:
SELECT attributs, fonction(attribut)
FROM tables GROUP BY ...
HAVING condition avec constante

squelette 5-2 : Agrégation conditionnée dynamique :
SELECT attributs, fonction(attribut)
FROM tables GROUP BY ...
HAVING condition avec sous-requête

squelette 1 - Projection :

SELECT attributs FROM table

SELECT * FROM emp;

empno	name	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800		20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30
7566	JONES	MANAGER	7839	1981-04-02	2975		20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30
7782	CLARK	MANAGER	7839	1981-06-09	2450		10
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20
7839	KING	PRESIDENT		1981-11-17	5000		10
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1983-01-12	1100		20
7900	JAMES	CLERK	7698	1981-12-03	950		30
7902	FORD	ANALYST	7566	1981-12-03	3000		20
7934	MILLER	CLERK	7782	1982-01-23	1300		10

squelette 1 - Projection :

SELECT attributs FROM table

SELECT name, hiredate, sal FROM emp;

empno	name	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800		20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30
7566	JONES	MANAGER	7839	1981-04-02	2975		20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30
7782	CLARK	MANAGER	7839	1981-06-09	2450		10
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20
7839	KING	PRESIDENT		1981-11-17	5000		10
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1983-01-12	1100		20
7900	JAMES	CLERK	7698	1981-12-03	950		30
7902	FORD	ANALYST	7566	1981-12-03	3000		20
7934	MILLER	CLERK	7782	1982-01-23	1300		10

squelette 1 - Projection :

SELECT attributs FROM table

SELECT name, hiredate, sal FROM emp;

name	hiredate	sal
SMITH	1980-12-17	800
ALLEN	1981-02-20	1600
WARD	1981-02-22	1250
JONES	1981-04-02	2975
MARTIN	1981-09-28	1250
BLAKE	1981-05-01	2850
CLARK	1981-06-09	2450
SCOTT	1982-12-09	3000
KING	1981-11-17	5000
TURNER	1981-09-08	1500
ADAMS	1983-01-12	1100
JAMES	1981-12-03	950
FORD	1981-12-03	3000
MILLER	1982-01-23	1300

squelette 2 - Implication de plusieurs entités

SELECT attributs FROM table1 JOIN table2

- 4 types de jointures :
 - jointure croisée : CROSS JOIN
 - produit cartésien des tables
 - jointure naturelle : NATURAL JOIN
 - jointure sur tous les champs portant le même nom
 - jointure interne : INNER JOIN
 - jointure sur tous les champs spécifiés dans le ON
 - jointure externe : OUTER JOIN
 - jointure permettant de récupérer des enregistrements sans correspondance entre les tables

squelette 2 - Implication de plusieurs entités : SELECT attributs FROM table1 JOIN table2

Jointure croisée : produit cartésien des tables

SELECT name, job, dname FROM emp CROSS JOIN dept;

empno	name	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800		20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30
7566	JONES	MANAGER	7839	1981-04-02	2975		20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30
7782	CLARK	MANAGER	7839	1981-06-09	2450		10
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20
7839	KING	PRESIDENT		1981-11-17	5000		10
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1983-01-12	1100		20
7900	JAMES	CLERK	7698	1981-12-03	950		30
7902	FORD	ANALYST	7566	1981-12-03	3000		20
7934	MILLER	CLERK	7782	1982-01-23	1300		10

deptno	dname	loc
10	ACC.	NY
20	RES.	DALLAS
30	SALES	CHICAGO
40	OPERAT.	BOSTON



squelette 2 - Implication de plusieurs entités : SELECT attributs FROM table1 JOIN table2

Jointure croisée : produit cartésien des tables

SELECT name, job, dname FROM emp CROSS JOIN dept;

name	job	dname
SMITH	CLERK	ACC.
SMITH	CLERK	RES.
SMITH	CLERK	SALES
SMITH	CLERK	OPERAT.
ALLEN	SALESMAN	ACC.
ALLEN	SALESMAN	RES.
ALLEN	SALESMAN	SALES
ALLEN	SALESMAN	OPERAT.



MILLER	CLERK	SALES
MILLER	CLERK	OPERAT.

squelette 2 - Implication de plusieurs entités : SELECT attributs FROM table1 JOIN table2

Jointure naturelle : jointure sur tous les champs avec le même nom
SELECT name, job, dname FROM emp NATURAL JOIN dept;

empno	name	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800		20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30
7566	JONES	MANAGER	7839	1981-04-02	2975		20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30
7782	CLARK	MANAGER	7839	1981-06-09	2450		10
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20
7839	KING	PRESIDENT		1981-11-17	5000		10
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1983-01-12	1100		20
7900	JAMES	CLERK	7698	1981-12-03	950		30
7902	FORD	ANALYST	7566	1981-12-03	3000		20
7934	MILLER	CLERK	7782	1982-01-23	1300		10

deptno	dname	loc
10	ACC.	NY
20	RES.	DALLAS
30	SALES	CHICAGO
40	OPERAT.	BOSTON



squelette 2 - Implication de plusieurs entités : SELECT attributs FROM table1 JOIN table2

Jointure naturelle : jointure sur tous les champs avec le même nom
SELECT name, job, dname FROM emp NATURAL JOIN dept;

name	job	dname
SMITH	CLERK	RESEARCH
ALLEN	SALESMAN	SALES
WARD	SALESMAN	SALES
JONES	MANAGER	RESEARCH
MARTIN	SALESMAN	SALES
BLAKE	MANAGER	SALES
CLARK	MANAGER	ACCOUNT.
SCOTT	ANALYST	RESEARCH
KING	PRESIDENT	ACCOUNT.
TURNER	SALESMAN	SALES
ADAMS	CLERK	RESEARCH
JAMES	CLERK	SALES
FORD	ANALYST	RESEARCH
MILLER	CLERK	ACCOUNT.

squelette 2 - Implication de plusieurs entités : SELECT attributs FROM table1 JOIN table2

Inner join : jointure sur tous les champs spécifiés dans le ON

SELECT name, job, dname FROM emp JOIN dept USING
(deptno) ;

SELECT name, job, dname FROM emp JOIN dept ON
dept.deptno=emp.deptno ;

empno	name	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800		20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30
7566	JONES	MANAGER	7839	1981-04-02	2975		20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30
7782	CLARK	MANAGER	7839	1981-06-09	2450		10
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20
7839	KING	PRESIDENT		1981-11-17	5000		10
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1983-01-12	1100		20
7900	JAMES	CLERK	7698	1981-12-03	950		30
7902	FORD	ANALYST	7566	1981-12-03	3000		20
7934	MILLER	CLERK	7782	1982-01-23	1300		10

...

deptno	dname	loc
10	ACC.	NY
20	RES.	DALLAS
30	SALES	CHICAGO
40	OPERAT.	BOSTON

squelette 2 - Implication de plusieurs entités : SELECT attributs FROM table1 JOIN table2

Inner join : jointure sur tous les champs spécifiés dans le ON
SELECT name, job, dname FROM emp JOIN dept USING
(deptno) ;

name	job	dname
SMITH	CLERK	RESEARCH
ALLEN	SALESMAN	SALES
WARD	SALESMAN	SALES
JONES	MANAGER	RESEARCH
MARTIN	SALESMAN	SALES
BLAKE	MANAGER	SALES
CLARK	MANAGER	ACCOUNT.
SCOTT	ANALYST	RESEARCH
KING	PRESIDENT	ACCOUNT.
TURNER	SALESMAN	SALES
ADAMS	CLERK	RESEARCH
JAMES	CLERK	SALES
FORD	ANALYST	RESEARCH
MILLER	CLERK	ACCOUNT.

squelette 2 - Implication de plusieurs entités : SELECT attributs FROM table1 JOIN table2

Outer join : récupération d'enregistrements sans correspondance entre les tables

SELECT name, job, dname FROM emp RIGHT OUTER JOIN dept USING (deptno);

empno	name	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800		20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30
7566	JONES	MANAGER	7839	1981-04-02	2975		20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30
7782	CLARK	MANAGER	7839	1981-06-09	2450		10
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20
7839	KING	PRESIDENT		1981-11-17	5000		10
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1983-01-12	1100		20
7900	JAMES	CLERK	7698	1981-12-03	950		30
7902	FORD	ANALYST	7566	1981-12-03	3000		20
7934	MILLER	CLERK	7782	1982-01-23	1300		10

deptno	dname	loc
10	ACC.	NY
20	RES.	DALLAS
30	SALES	CHICAGO
40	OPERAT.	BOSTON

squelette 2 - Implication de plusieurs entités : SELECT attributs FROM table1 JOIN table2

Outer join : récupération d'enregistrements sans
correspondance entre les tables

SELECT name, job, dname FROM emp RIGHT OUTER JOIN
dept USING (deptno);

name	job	dname
CLARK	MANAGER	ACCOUNT.
KING	PRESIDENT	ACCOUNT.
MILLER	CLERK	ACCOUNT.
SMITH	CLERK	RESEARCH
JONES	MANAGER	RESEARCH
SCOTT	ANALYST	RESEARCH
ADAMS	CLERK	RESEARCH
FORD	ANALYST	RESEARCH
ALLEN	SALESMAN	SALES
WARD	SALESMAN	SALES
MARTIN	SALESMAN	SALES
BLAKE	MANAGER	SALES
TURNER	SALESMAN	SALES
JAMES	CLERK	SALES
		OPERAT.

squelette 2 - Implication de plusieurs entités : SELECT attributs FROM table1 JOIN table2

Outer join : récupération d'enregistrements sans
correspondance entre les tables

SELECT name, job, dname FROM emp LEFT OUTER JOIN
dept USING (deptno);

name	job	dname
CLARK	MANAGER	ACCOUNT.
KING	PRESIDENT	ACCOUNT.
MILLER	CLERK	ACCOUNT.
SMITH	CLERK	RESEARCH
JONES	MANAGER	RESEARCH
SCOTT	ANALYST	RESEARCH
ADAMS	CLERK	RESEARCH
FORD	ANALYST	RESEARCH
ALLEN	SALESMAN	SALES
WARD	SALESMAN	SALES
MARTIN	SALESMAN	SALES
BLAKE	MANAGER	SALES
TURNER	SALESMAN	SALES
JAMES	CLERK	SALES

squelette 3-1 - Restriction simple :

SELECT attributs FROM tables WHERE conditions

SELECT * FROM emp WHERE sal>=1500;

empno	name	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800		20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30
7566	JONES	MANAGER	7839	1981-04-02	2975		20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30
7782	CLARK	MANAGER	7839	1981-06-09	2450		10
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20
7839	KING	PRESIDENT		1981-11-17	5000		10
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1983-01-12	1100		20
7900	JAMES	CLERK	7698	1981-12-03	950		30
7902	FORD	ANALYST	7566	1981-12-03	3000		20
7934	MILLER	CLERK	7782	1982-01-23	1300		10

squelette 3-1 - Restriction simple :

SELECT attributs FROM tables WHERE conditions

SELECT * FROM emp WHERE sal>=1500;

empno	name	job	mgr	hiredate	sal	comm	deptno
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7566	JONES	MANAGER	7839	1981-04-02	2975		20
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30
7782	CLARK	MANAGER	7839	1981-06-09	2450		10
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20
7839	KING	PRESIDENT		1981-11-17	5000		10
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7902	FORD	ANALYST	7566	1981-12-03	3000		20

squelette 3-1 - Restriction simple & projection: SELECT attributs FROM tables WHERE conditions

SELECT name, hiredate, sal FROM emp WHERE sal>=1500;

empno	name	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800		20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30
7566	JONES	MANAGER	7839	1981-04-02	2975		20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30
7782	CLARK	MANAGER	7839	1981-06-09	2450		10
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20
7839	KING	PRESIDENT		1981-11-17	5000		10
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1983-01-12	1100		20
7900	JAMES	CLERK	7698	1981-12-03	950		30
7902	FORD	ANALYST	7566	1981-12-03	3000		20
7934	MILLER	CLERK	7782	1982-01-23	1300		10

squelette 3-1 - Restriction simple & projection:

SELECT attributs FROM tables WHERE conditions

SELECT name, hiredate, sal FROM emp WHERE sal >= 1500;

name	hiredate	sal
ALLEN	1981-02-20	1600
JONES	1981-04-02	2975
BLAKE	1981-05-01	2850
CLARK	1981-06-09	2450
SCOTT	1982-12-09	3000
KING	1981-11-17	5000
TURNER	1981-09-08	1500
FORD	1981-12-03	3000

squelette 4 – Fct. de groupes ou d'agrégation

SELECT attributs, fonction(attribut) FROM tables
GROUP BY ...

SELECT job, avg(sal) FROM emp GROUP BY job;

empno	name	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800		20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30
7566	JONES	MANAGER	7839	1981-04-02	2975		20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30
7782	CLARK	MANAGER	7839	1981-06-09	2450		10
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20
7839	KING	PRESIDENT		1981-11-17	5000		10
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1983-01-12	1100		20
7900	JAMES	CLERK	7698	1981-12-03	950		30
7902	FORD	ANALYST	7566	1981-12-03	3000		20
7934	MILLER	CLERK	7782	1982-01-23	1300		10

squelette 4 – Fct. de groupes ou d'agrégation

SELECT attributs, fonction(attribut) FROM tables
GROUP BY ...

SELECT job, avg(sal) FROM emp GROUP BY job;

job	avg(sal)
SALESMAN	1400,00
MANAGER	2758,33
CLERK	1037,50
PRESIDENT	5000,00
ANALYST	3000,00

- ATTENTION : **UNE** seule valeur retournée par groupe

SELECT ~~name~~, job, avg(sal) FROM emp GROUP BY job;

plusieurs valeurs de « name » pour chaque groupe !!!

squelette 4 - Fct. de groupes & restriction :

SELECT attributs, fonction(attribut) FROM tables
GROUP BY ...

SELECT job, avg(sal) FROM emp WHERE name like '%A%'
GROUP BY job;

empno	name	job	mgr	hiredate	sal	comm	deptno
7369	SMITH	CLERK	7902	1980-12-17	800		20
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30
7566	JONES	MANAGER	7839	1981-04-02	2975		20
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30
7782	CLARK	MANAGER	7839	1981-06-09	2450		10
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20
7839	KING	PRESIDENT		1981-11-17	5000		10
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30
7876	ADAMS	CLERK	7788	1983-01-12	1100		20
7900	JAMES	CLERK	7698	1981-12-03	950		30
7902	FORD	ANALYST	7566	1981-12-03	3000		20
7934	MILLER	CLERK	7782	1982-01-23	1300		10

squelette 4 - Fct. de groupes & restriction :

SELECT attributs, fonction(attribut) FROM tables
GROUP BY ...

SELECT job, avg(sal) FROM emp WHERE name like '%A%'
GROUP BY job;

job	avg(sal)
SALESMAN	1366,67
MANAGER	2650,00
CLERK	1025,00

- ATTENTION : Toujours **UNE** seule valeur retournée par groupe

Patron 5-1 - Agrégation conditionnée par une cste: SELECT attributs, fonction(attribut) FROM tables GROUP BY ... HAVING condition avec constante

SELECT job, avg(sal) FROM emp GROUP BY job HAVING
avg(sal)>1300;

empno	name	job	mgr	hiredate	sal	comm	deptno	avg(sal)
7369	SMITH	CLERK	7902	1980-12-17	800		20	1037,50
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30	1400,00
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30	1400,00
7566	JONES	MANAGER	7839	1981-04-02	2975		20	2758,33
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30	1400,00
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30	2758,33
7782	CLARK	MANAGER	7839	1981-06-09	2450		10	2758,33
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20	3000,00
7839	KING	PRESIDENT		1981-11-17	5000		10	5000,00
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30	1400,00
7876	ADAMS	CLERK	7788	1983-01-12	1100		20	1037,50
7900	JAMES	CLERK	7698	1981-12-03	950		30	1037,50
7902	FORD	ANALYST	7566	1981-12-03	3000		20	3000,00
7934	MILLER	CLERK	7782	1982-01-23	1300		10	1037,50

squelette 5-1 - Agrégation conditionnée par une cste: SELECT attributs, fonction(attribut) FROM tables GROUP BY ... HAVING condition avec constante

SELECT job, avg(sal) FROM emp GROUP BY job HAVING avg(sal)>1300;

job	avg(sal)
SALESMAN	1400,00
MANAGER	2758,33
CLERK	1037,50
PRESIDENT	5000,00
ANALYST	3000,00

avg(sal)>1300

job	avg(sal)
SALESMAN	1400,00
MANAGER	2758,33
PRESIDENT	5000,00
ANALYST	3000,00

squelette 5.2 - Agrégation conditionnée dyn.: SELECT attributs, fonction(attribut) FROM tables GROUP BY ... HAVING condition avec sous-requête

SELECT job, avg(sal) FROM emp GROUP BY job HAVING
avg(sal)>=ALL(SELECT avg(sal) FROM emp GROUP BY job) ;

empno	name	job	mgr	hiredate	sal	comm	deptno	avg(sal)
7369	SMITH	CLERK	7902	1980-12-17	800		20	1037,50
7499	ALLEN	SALESMAN	7698	1981-02-20	1600	300	30	1400,00
7521	WARD	SALESMAN	7698	1981-02-22	1250	500	30	1400,00
7566	JONES	MANAGER	7839	1981-04-02	2975		20	2758,33
7654	MARTIN	SALESMAN	7698	1981-09-28	1250	1400	30	1400,00
7698	BLAKE	MANAGER	7839	1981-05-01	2850		30	2758,33
7782	CLARK	MANAGER	7839	1981-06-09	2450		10	2758,33
7788	SCOTT	ANALYST	7566	1982-12-09	3000		20	3000,00
7839	KING	PRESIDENT		1981-11-17	5000		10	5000,00
7844	TURNER	SALESMAN	7698	1981-09-08	1500	0	30	1400,00
7876	ADAMS	CLERK	7788	1983-01-12	1100		20	1037,50
7900	JAMES	CLERK	7698	1981-12-03	950		30	1037,50
7902	FORD	ANALYST	7566	1981-12-03	3000		20	3000,00
7934	MILLER	CLERK	7782	1982-01-23	1300		10	1037,50

squelette 5.2 - Agrégation conditionnée dyn.:

SELECT attributs, fonction(attribut) FROM tables
GROUP BY ... HAVING condition avec sous-requête

SELECT job, avg(sal) FROM emp GROUP BY job HAVING
avg(sal) >= ALL(SELECT avg(sal) FROM emp GROUP BY job) ;

job	avg(sal)
PRESIDENT	5000,00